

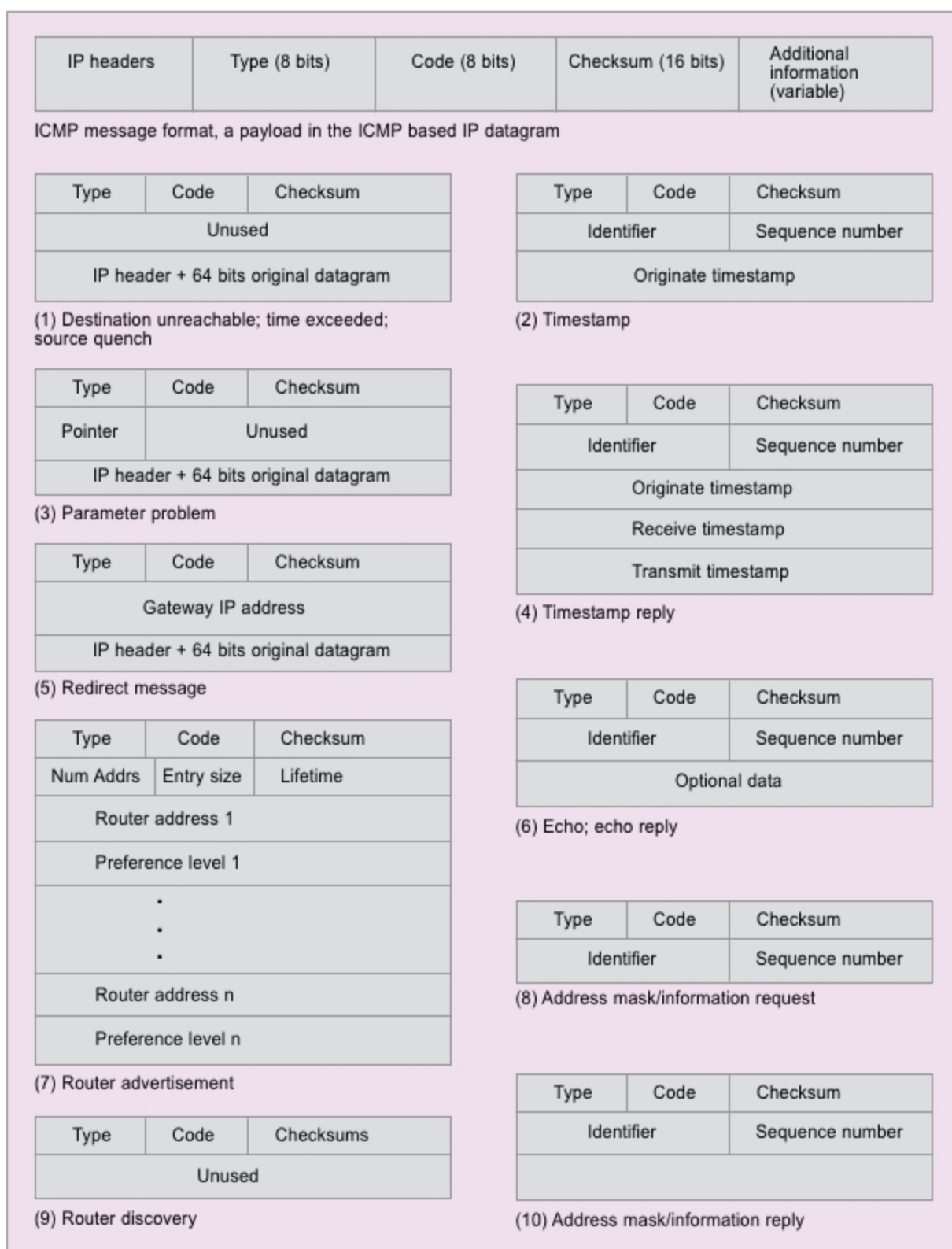


Fig. 2: Illustrations of various ICMP message formats

other tasks. A few examples of ICMP messages are:

- ICMP type 0, code 0 is used to ping with 16-bit identifier and sequence number.
- ICMP type 3, code 0 is used to indicate host unreachable; codes 1, 2, 3, 4 and 5, respectively, indicate the statuses of host unreachable, protocol unreachable, port unreachable, fragmentation required and source route failure, among others.
- ICMP type 4, code 0 is used for congestion control required/destination unreachable status.
- ICMP type 5, code 0 is used for network error; code 1 for host, code 2 for type of service and network, code 3 for type of service and host redirect

message, 32-bit gateway address Internet Header, 64-bit original payload.

- ICMP type 8, code 0 is for echo request/reply, 16-bit identifier, 16-bit sequence number, optional data.
- ICMP type 11, code 0 is for time-to-live when expired on transit; code 1 is used when fragmentation reassembly time is exceeded, expiry of TTL, 32 bits unused, Internet header, 64 bits of original payload.
- ICMP type 12, code 0 indicates error parameter problem, 8-bit pointer, rest 24 bits unused, Internet header, 64-bit original payload.
- ICMP type 13, code 0 is used for timestamp request with 16-bit identifier, 16-bit sequence number, 32-bit original timestamp.

- ICMP type 14, code 0 is for timestamp reply with 16-bit identifier, 16-bit sequence number, 32-bit original timestamp, 32-bit receive timestamp, 32-bit transit timestamp.

- ICMP type 15, code 0 is used for information request for address mask with 16-bit identifier, 16-bit sequence number.

- ICMP type 16, code is 0 used for information reply for address mask.

An ICMP message is made of a type field, a code field and the first eight bytes of an IP datagram (Fig. 2). A datagram is for the ICMP message generated in the first place so that the sender knows about the packet that caused the error. An ICMP message is sent as IP datagram with TOS field set to 0 and protocol field set to 1 that defines the ICMP pack.

After headers, ICMP message is included as payload. ICMP message format has four parts: 1-byte type field, 1-byte code field, 2-byte checksum field used to provide checksum (1s complement addition) of all 16-bit headers and additional information field, depending on the header of the ICMP message.

For timestamp request, additional information is made of 2-byte identifier, 2-byte sequence number and 4-byte originating timestamp. For echo request and reply, additional information is made of 1-byte identifier, 1-byte sequence number and original IP header. For destination not reachable and others, additional information contains 4 bytes of unused fields, followed by the original IP header.

A new version of ICMP has been defined for IPv6. It does use existing types and codes, but has added new types and codes. Added new ICMP type includes Packet Too Big, Unrecognised IPv6 options and others. Different error-reporting ICMP messages are Destination Unreachable, Source Quench, Time Exceeded, Parameter Problem and Redirect. Various ICMP messages for management query are echo request and reply, timestamp request and reply, address mask request and reply, and router request and advertisement. **EFY**

This article is dedicated to Internet Engineering Task Force (IETF) for proposing IPv6 to provide global accessibility of the Internet